Real-Time Speech Noise Reduction Using Digital Filters: A Causal versus Offline Comparison of Butterworth and Notch Filters

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***Abstract*—Digital filters remain the lowest-cost route to suppressing narrowband and band-limited noise in speech, yet reported gains are often measured with offline zero-phase filtering that cannot run in real time. This paper compares four filter families, Butterworth low-pass (LPF), high-pass (HPF), band-pass (BPF) and IIR notch filters, on real speech corrupted with synthetic white, mains-hum, rumble and hiss noise. Each filter is evaluated both as a true causal, frame-by-frame streaming filter and as an offline forward–backward (zero-phase) filter, using SNR, SNR improvement (ΔSNR) and mean squared error (MSE). Over 6 clips, 7 noise conditions and 10 noise realisations per cell, the causal LPF gives $+2.89$ dB ΔSNR for 5 dB white noise and $+11.46$ dB for high-frequency hiss, while a three-harmonic notch bank gives $+11.49$ dB against 50 Hz hum (causal) and $+16.70$ dB (zero-phase). Causal HPF and BPF show negative waveform ΔSNR despite removing noise power, because their group delay of about 6 ms near 100 Hz distorts the waveform; a phase-insensitive output-referred metric and a zero-phase evaluation expose this effect. Streaming output is bit-identical to batch causal filtering. All code, data and results are provided.**

***Index Terms*—speech enhancement, noise reduction, Butterworth filter, notch filter, real-time processing, causal filtering, zero-phase filtering**

I. Introduction

Background noise degrades the quality and intelligibility of speech in telephony, hearing aids and voice interfaces. Modern deep-learning enhancers have advanced rapidly [1]–[3], but classical digital filters remain attractive on resource-constrained hardware because they need no training data, have a few multiply–accumulates per sample, and have analytically known latency. Filters are also the right tool when the interference has a known spectral location, such as mains hum at 50 or 60 Hz, low-frequency rumble, or out-of-band hiss.

A practical issue is often overlooked: many filter evaluations use zero-phase (forward–backward) filtering, which needs future samples and is therefore not causal. A real-time system must use a causal filter, whose phase response introduces frequency-dependent delay. This paper therefore asks two questions: (i) which of LPF, HPF, BPF and notch filters best removes which type of noise, and (ii) how much of the benefit survives when the filter is run causally in a frame-by-frame streaming implementation?

The contributions are: a reproducible comparison of five Butterworth/notch filter configurations under seven noise conditions; an explicit causal-versus-offline analysis with delay compensation; a streaming implementation verified to equal batch causal filtering; and a complete MATLAB/Octave and Python code base with the speech data used.

II. Related Work

Supervised deep-learning methods now dominate single-channel speech enhancement; Wang and Chen survey the field [1], and Das *et al.* and Michelsanti *et al.* give broader and audio-visual overviews [2], [3]. Early neural approaches include WaveNet-style waveform denoising [4]. Recent surveys cover generative adversarial approaches [5] and diffusion models for audio restoration [6].

Real-time operation has been a focus of recent work. Valin combined DSP and deep learning for full-band enhancement [7]. The Interspeech 2020 Deep Noise Suppression Challenge provided datasets and a subjective testing framework for real-time single-channel enhancement and observed that common objective metrics correlate poorly with subjective tests [8]. Entries and later models include a causal waveform-domain model that runs in real time on a laptop CPU [9], the complex-valued DCCRN [10], the small DTLN [11], FullSubNet [12], and the low-complexity DeepFilterNet [13] with its successors, which report real-time factors of 0.04 on a notebook CPU for embedded-oriented operation [14] and 0.19 on a single thread for a perceptually motivated version [15]. These systems target broadband, non-stationary noise and need training data and considerably more computation than a biquad cascade. The present paper addresses the complementary, well-defined case in which the noise has a known spectral signature and a classical filter suffices, and it quantifies what causality costs.

III. Methodology

A. Signal model and metrics

A clean speech signal $s[n]$ is corrupted by additive noise $n[n]$:

$$x[n] = s[n] + n[n], \qquad \mathrm{SNR}_{\mathrm{in}} = 10\log_{10}\frac{\sum_n s^2[n]}{\sum_n n^2[n]}. \quad (1)$$

TABLE I
FILTER CONFIGURATIONS ($f_s = 16$ kHz).

Name	Design	Parameters
LPF	Butterworth low-pass	$N = 4$, $f_c = 3400$ Hz
HPF	Butterworth high-pass	$N = 4$, $f_c = 100$ Hz
BPF	Butterworth band-pass	$N = 4$ (8th order), 100–3400 Hz
Notch	Single IIR notch	$f_0 = 50$ Hz, $Q = 30$
Notch-H	Notch bank	$f_0 \in \{50, 100, 150\}$ Hz, $Q = 30$

The noise is scaled so that $\mathrm{SNR}_{\mathrm{in}}$ equals its target exactly for every clip. A filter H produces $\hat{s} = H\{x\}$, and the waveform metrics against the clean reference are

$$\mathrm{SNR}_{\mathrm{out}} = 10 \log_{10} \frac{\sum_n s^2[n]}{\sum_n (s[n] - \hat{s}[n])^2}, \tag{2}$$

$$\Delta\mathrm{SNR} = \mathrm{SNR}_{\mathrm{out}} - \mathrm{SNR}_{\mathrm{in}}, \tag{3}$$

$$\mathrm{MSE} = \frac{1}{N} \sum_{n=0}^{N-1} \big(s[n] - \hat{s}[n]\big)^2. \tag{4}$$

Because (2) compares waveforms, it penalises both residual noise and any speech distortion or delay introduced by the filter. Since all filters here are linear, $\hat{s} = H\{s\} + H\{n\}$, which allows a complementary output-referred SNR that is independent of filter delay and phase:

$$\mathrm{SNR}_{\mathrm{oref}} = 10 \log_{10} \frac{\sum_n (H\{s\}[n])^2}{\sum_n (H\{n\}[n])^2}. \tag{5}$$

A third, phase-insensitive metric applies (2) to STFT magnitudes (512-point Hann window, 50% overlap). All clips are scaled to RMS 0.1, so MSE values are comparable across clips and $\mathrm{MSE}_{\mathrm{in}} = 0.01 \cdot 10^{-\mathrm{SNR}_{\mathrm{in}}/10}$.

B. Filters

The Butterworth magnitude response of order N and cut-off f_c is

$$|H_{\mathrm{LP}}(f)|^2 = \frac{1}{1 + (f/f_c)^{2N}}, \qquad |H_{\mathrm{HP}}(f)|^2 = \frac{1}{1 + (f_c/f)^{2N}}. \tag{6}$$

Digital filters are obtained by the bilinear transform with pre-warping, and are implemented as cascaded second-order sections (biquads) in transposed direct form II,

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2] - a_1 y[n-1] - a_2 y[n-2], \tag{7}$$

which is numerically robust for the 100 Hz corner at $f_s = 16$ kHz. The notch filter at f_0 with quality factor Q is

$$H_{\mathrm{N}}(z) = g \, \frac{1 - 2\cos\omega_0 z^{-1} + z^{-2}}{1 - 2g\cos\omega_0 z^{-1} + (2g - 1) z^{-2}}, \tag{8}$$

with $\omega_0 = 2\pi f_0 / f_s$, $\beta = \tan(\omega_0/(2Q))$ and $g = 1/(1 + \beta)$. Table I lists the five configurations.

C. Causal (real-time) versus offline filtering

A causal filter computes $y[n]$ from $x[m]$, $m \le n$, so it can run sample-by-sample or frame-by-frame with state carried across frames, and its only inherent delay is its group delay

$$\tau_g(\omega) = -\frac{d \angle H(e^{j\omega})}{d\omega}. \tag{9}$$

Butterworth filters are minimum-phase-like but not linear-phase, so τ_g varies with frequency (Fig. 3c). The total algorithmic latency of a streaming implementation is the frame-buffering time plus τ_g.

Offline zero-phase filtering runs the filter forward, reverses the output, runs it again and reverses it back:

$$y_z = \mathcal{R}\{H\{\mathcal{R}\{H\{x\}\}\}\}, \qquad |H_z(f)| = |H(f)|^2, \quad \angle H_z = 0, \tag{10}$$

where $\mathcal{R}$ reverses the sequence. This requires the whole signal (future samples), so it is *not* real-time. It doubles the attenuation in dB and moves the -3 dB corner to -6 dB (Fig. 3b).

For the causal outputs two evaluations are reported. *Raw* compares $\hat{s}[n]$ with $s[n]$ directly. *Delay-compensated* advances $\hat{s}$ by the integer lag $\tau \in [0, 160]$ samples that maximises the cross-correlation between s and $H\{s\}$, a bulk-delay correction that is available only offline but shows how much error is due to a pure delay. A single integer lag cannot remove frequency-dependent delay, which is the central finding below.

D. Experimental setup

Speech. Six real utterances (19.4 s in total) from the CMU Arctic corpus (two speakers, 16 kHz, mono) were taken from the pyroomacoustics repository example data. DC was removed and each clip scaled to RMS 0.1.

Noise. Seven conditions were synthesised with seeded generators: white Gaussian noise (WGN) at 0, 5 and 10 dB; mains hum with components at 50, 100 and 150 Hz (relative amplitudes 1, 0.5, 0.25, random phases) at 5 dB; rumble (20–80 Hz band-limited noise) at 5 dB; hiss (5–7.9 kHz band-limited noise) at 5 dB; and hum plus WGN of equal power at 5 dB total.

Trials. Each condition–filter–mode cell uses 6 clips $\times$ 10 noise seeds $= 60$ trials; tables give mean $\pm$ standard deviation.

Streaming. The causal filters were also run frame-by-frame (5, 10, 20 and 32 ms frames) with state carried between frames and compared with batch causal output; per-frame processing time was measured.

Software. The main results were produced with Python (NumPy 2.4.4, SciPy 1.17.1). The MATLAB implementation was executed in GNU Octave 8.4.0 (not in MATLAB) and agreed with the Python results (Section IV-F). Fig. 1 shows the system and Fig. 2 the experimental procedure.

IV. RESULTS

A. Waveforms and spectra

Fig. 4 shows one utterance in 5 dB white noise and the causal LPF and BPF outputs. Fig. 5 (top left) shows that the white noise floor dominates the speech spectrum above about 3 kHz, which the LPF and BPF remove; Fig. 5 (top right) shows that the 50 Hz hum peak is reduced by the single notch and further, together with the 100 and 150 Hz harmonics, by the notch bank. The spectrograms in the lower half of Fig. 5 show the same behaviour in the time–frequency plane.

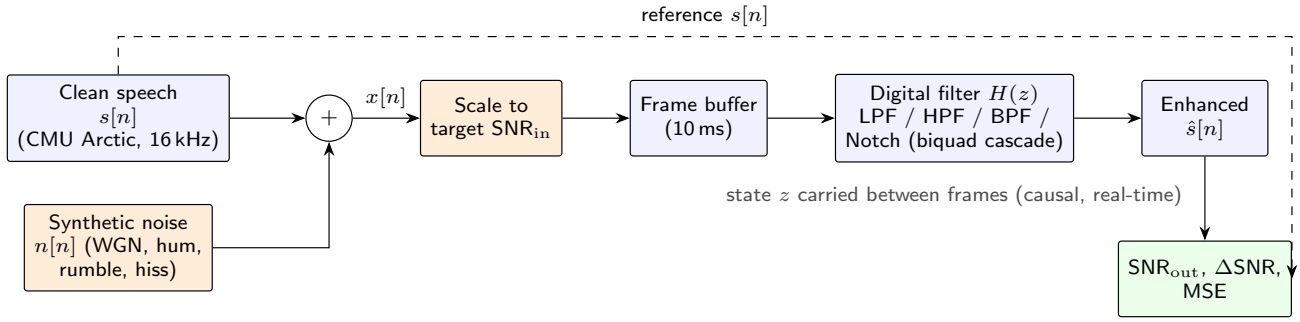


Fig. 1. System block diagram of the evaluation chain with frame-based causal filtering.

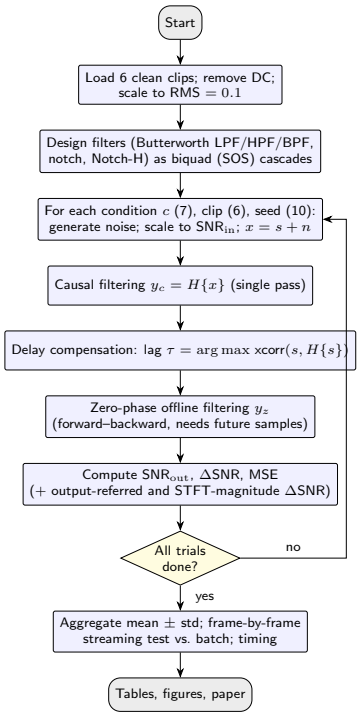


Fig. 2. Methodology flowchart.

B. Waveform SNR improvement and MSE

Tables II and III give the real-time (causal) results and Table IV the offline zero-phase ΔSNR; Fig. 6 summarises the causal case, and the zero-phase MSE is shown in the lower half of Fig. 7 (all values are in `results/metrics_summary.csv`).

C. Causal versus offline filtering

Fig. 7 compares raw causal, delay-compensated causal and zero-phase results for the BPF and LPF. Delay compensation matters for the LPF (e.g. -0.73 dB raw versus $+2.89$ dB compensated for 5 dB WGN) because the LPF delay is nearly constant (≈ 0.1 ms) and a 2-sample shift removes most of it. It does not rescue the BPF and HPF, whose delay varies from about 6 ms at 100 Hz to under 0.3 ms above 500 Hz.

D. Delay- and phase-independent metrics

Table V reports the output-referred ΔSNR of (5). For white noise, the LPF and BPF give 3.6 dB at every input SNR, close to the ideal $10\log_{10}(8000/3400) = 3.72$ dB expected from removing the noise above 3.4 kHz from a flat spectrum; the HPF and notches remove almost no white-noise power. For narrowband noise the picture is clear: the HPF and BPF give 13.6 dB against rumble, the LPF and BPF 25.2–25.5 dB against hiss, and the notch bank 15.6 dB against hum (21.6 dB zero-phase). The STFT-magnitude ΔSNR (available in `results/`) shows the same qualitative picture and, unlike the waveform metric, is positive for the causal HPF and BPF against rumble and hum (e.g. $+13.0$ dB for HPF against rumble, causal).

E. Real-time behaviour

Table VI and Fig. 8 give group delay and streaming measurements. Streaming output was identical to batch causal output for every filter and frame length (maximum absolute difference 0). In the authors' Python sandbox the mean time per 10 ms frame was 0.03–0.04 ms (real-time factor about 0.003); absolute timings depend on hardware and language and are not claims about embedded targets. With a 10 ms frame, the latency of the LPF, notch and notch bank is dominated by buffering, whereas HPF/BPF add about 6 ms of group delay near 100 Hz.

F. Validation of the MATLAB implementation

On a deterministic test (hum with fixed phases, one clip) the Octave-executed MATLAB code and the Python code agreed to within 5×10^{-7} dB in SNR and the ΔSNR variants and to 2.5×10^{-11} in MSE (difference attributable to CSV rounding). On the full randomised experiment, which uses different random-number generators, the mean absolute difference in ΔSNR was 0.004 dB and the maximum 0.11 dB.

V. Discussion

Matching the filter to the noise. No single filter is best. The LPF is the best causal choice against white and hiss noise; the notch bank is best against hum; and the HPF is the right tool for rumble, though only the zero-phase version scores well on waveform SNR (Table IV). A single 50 Hz notch

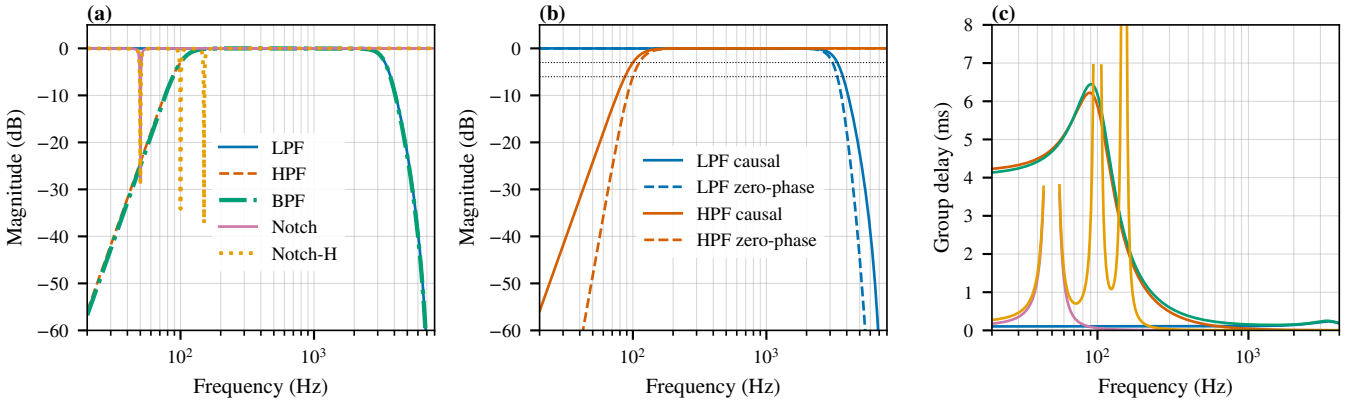


Fig. 3. Filter responses. (a) Magnitude responses of the five filters. (b) Causal versus zero-phase magnitude of LPF and HPF (dotted lines: -3 and -6 dB). (c) Group delay; spikes near notch centres are where group delay is undefined.

TABLE II
WAVEFORM ΔSNR (DB), REAL-TIME CAUSAL FILTERING WITH INTEGER DELAY COMPENSATION (MEAN $\pm$ STD OVER 60 TRIALS; BEST PER ROW IN BOLD).

Condition	LPF	HPF	BPF	Notch	Notch-H
WGN 0 dB	**3.41 ± 0.23**	−2.74 ± 0.43	−1.04 ± 0.73	0.00 ± 0.00	−0.06 ± 0.06
WGN 5 dB	**2.89 ± 0.61**	−5.79 ± 0.69	−4.93 ± 0.95	0.00 ± 0.00	−0.21 ± 0.19
WGN 10 dB	**1.62 ± 1.33**	−9.92 ± 0.84	−9.50 ± 1.04	0.00 ± 0.00	−0.63 ± 0.53
Hum 5 dB	−0.38 ± 0.30	−4.67 ± 0.89	−4.52 ± 1.03	5.80 ± 0.16	**11.49 ± 2.36**
Rumble 5 dB	−0.38 ± 0.30	−4.51 ± 0.92	−4.35 ± 1.08	**0.19 ± 0.05**	−0.01 ± 0.21
Hiss 5 dB	**11.46 ± 3.34**	−5.80 ± 0.68	−4.29 ± 1.09	0.00 ± 0.00	−0.22 ± 0.19
Hum+WGN 5 dB	0.95 ± 0.40	−5.26 ± 0.77	−4.72 ± 0.98	2.00 ± 0.04	**2.48 ± 0.32**

TABLE III
MSE AGAINST THE CLEAN REFERENCE ($\times 10^{-3}$, CLIPS NORMALISED TO RMS 0.1), CAUSAL FILTERING WITH DELAY COMPENSATION (MEAN $\pm$ STD; LOWEST PER ROW IN BOLD).

Condition	Unfilt.	LPF	HPF	BPF	Notch	Notch-H
WGN 0 dB	10.00	**4.56 ± 0.25**	18.87 ± 1.86	12.89 ± 2.08	10.00 ± 0.00	10.15 ± 0.15
WGN 5 dB	3.16	**1.64 ± 0.24**	12.13 ± 1.87	10.06 ± 2.09	3.16 ± 0.00	3.32 ± 0.14
WGN 10 dB	1.00	**0.72 ± 0.24**	9.99 ± 1.86	9.16 ± 2.08	1.00 ± 0.00	1.16 ± 0.14
Hum 5 dB	3.16	3.46 ± 0.24	9.45 ± 1.86	9.19 ± 2.08	0.83 ± 0.03	**0.26 ± 0.12**
Rumble 5 dB	3.16	3.46 ± 0.24	9.14 ± 1.86	8.87 ± 2.08	**3.03 ± 0.03**	3.17 ± 0.15
Hiss 5 dB	3.16	**0.30 ± 0.24**	12.17 ± 1.86	8.75 ± 2.08	3.16 ± 0.00	3.33 ± 0.14
Hum+WGN 5 dB	3.16	2.55 ± 0.24	10.77 ± 1.85	9.60 ± 2.06	2.00 ± 0.02	**1.79 ± 0.13**

TABLE IV
WAVEFORM ΔSNR (DB), OFFLINE ZERO-PHASE (FORWARD–BACKWARD) FILTERING (MEAN $\pm$ STD OVER 60 TRIALS; BEST PER ROW IN BOLD).

Condition	LPF	HPF	BPF	Notch	Notch-H
WGN 0 dB	3.81 ± 0.27	0.02 ± 0.03	**3.84 ± 0.34**	0.00 ± 0.00	−0.02 ± 0.04
WGN 5 dB	**3.28 ± 0.72**	−0.06 ± 0.11	3.15 ± 0.87	0.00 ± 0.00	−0.09 ± 0.11
WGN 10 dB	**2.04 ± 1.56**	−0.32 ± 0.32	1.64 ± 1.83	0.00 ± 0.00	−0.31 ± 0.34
Hum 5 dB	−0.36 ± 0.32	9.25 ± 0.94	7.27 ± 2.09	6.12 ± 0.04	**16.70 ± 4.24**
Rumble 5 dB	−0.36 ± 0.32	**14.88 ± 3.20**	10.48 ± 3.88	0.28 ± 0.06	0.22 ± 0.15
Hiss 5 dB	**12.51 ± 4.27**	−0.12 ± 0.11	11.38 ± 4.59	0.00 ± 0.00	−0.11 ± 0.11
Hum+WGN 5 dB	1.09 ± 0.44	2.45 ± 0.20	**4.70 ± 1.23**	2.06 ± 0.02	2.79 ± 0.21

removes only the fundamental of the hum (+5.80 dB) and the three-notch bank doubles the benefit (+11.49 dB) because the fundamental carries only about 76% of the hum power in this model (amplitudes 1, 0.5, 0.25). Against broadband white noise, notches and the HPF do nothing useful (HPF causal ΔSNR -2.74 to -9.92 dB), and the benefit of LPF/BPF is bounded by the bandwidth argument above, so classical filters are not a substitute for learned enhancers [9], [10], [13] when noise overlaps speech in frequency.

Why causality costs waveform SNR. The causal HPF and BPF have negative waveform ΔSNR in every condition (Table II), including those where the output-referred metric shows large noise removal (Table V). The group delay peaks near the 100 Hz corner (Fig. 3c), exactly where voiced speech

TABLE V

OUTPUT-REFERRED ΔSNR (DB), $10\log_{10}(\|Hs\|^2/\|Hn\|^2) - \mathrm{SNR}_{\mathrm{in}}$, CAUSAL FILTERING. THIS METRIC IS INDEPENDENT OF FILTER DELAY AND PHASE AND ISOLATES NOISE-POWER REMOVAL.

Condition	LPF	HPF	BPF	Notch	Notch-H
WGN 0 dB	3.56 ± 0.13	-0.04 ± 0.08	$\mathbf{3.59 \pm 0.20}$	0.00 ± 0.00	-0.06 ± 0.07
WGN 5 dB	3.57 ± 0.13	-0.04 ± 0.08	$\mathbf{3.60 \pm 0.20}$	0.00 ± 0.00	-0.06 ± 0.07
WGN 10 dB	3.55 ± 0.13	-0.04 ± 0.08	$\mathbf{3.58 \pm 0.20}$	0.00 ± 0.00	-0.06 ± 0.07
Hum 5 dB	-0.14 ± 0.12	8.32 ± 0.08	8.18 ± 0.20	5.80 ± 0.16	$\mathbf{15.62 \pm 1.38}$
Rumble 5 dB	-0.14 ± 0.12	13.61 ± 0.31	$\mathbf{13.68 \pm 0.34}$	0.19 ± 0.05	0.15 ± 0.09
Hiss 5 dB	25.24 ± 0.14	-0.10 ± 0.08	$\mathbf{25.50 \pm 0.20}$	0.00 ± 0.00	-0.07 ± 0.07
Hum+WGN 5 dB	1.33 ± 0.12	2.38 ± 0.09	$\mathbf{5.31 \pm 0.20}$	2.00 ± 0.04	2.82 ± 0.06

TABLE VI

REAL-TIME CHARACTERISTICS OF THE CAUSAL FILTERS AT $f_s = 16$ KHZ (FRAME-BY-FRAME STREAMING, 10 MS FRAMES). GROUP DELAY (GD) FROM THE ANALYTIC PHASE RESPONSE; "MAX GD IS THE MAXIMUM OVER 100–3400 HZ (EXCLUDING $\pm$10 HZ AROUND NOTCH CENTRES, WHERE GD IS UNDEFINED). TIMING WAS MEASURED ON THE AUTHORS' SANDBOX CPU WITH SCIPY AND WILL DIFFER ON OTHER HARDWARE.

Filter	Order	GD@200 Hz (ms)	GD@1 kHz (ms)	max GD (ms)	est. lag samples (ms)	mean time (ms/frame)	max time (ms/frame)	RTF	max|stream−batch|
LPF	4	0.10	0.11	0.24	2.0 (0.12)	0.039	0.133	0.0039	0.0e+00
HPF	4	1.19	0.04	5.85	10.3 (0.65)	0.032	0.099	0.0032	0.0e+00
BPF	8	1.31	0.16	6.16	10.7 (0.67)	0.035	0.125	0.0035	0.0e+00
Notch	2	0.01	0.00	0.06	0.0 (0.00)	0.032	0.098	0.0032	0.0e+00
Notch-H	6	0.20	0.00	3.96	0.0 (0.00)	0.033	0.090	0.0033	0.0e+00

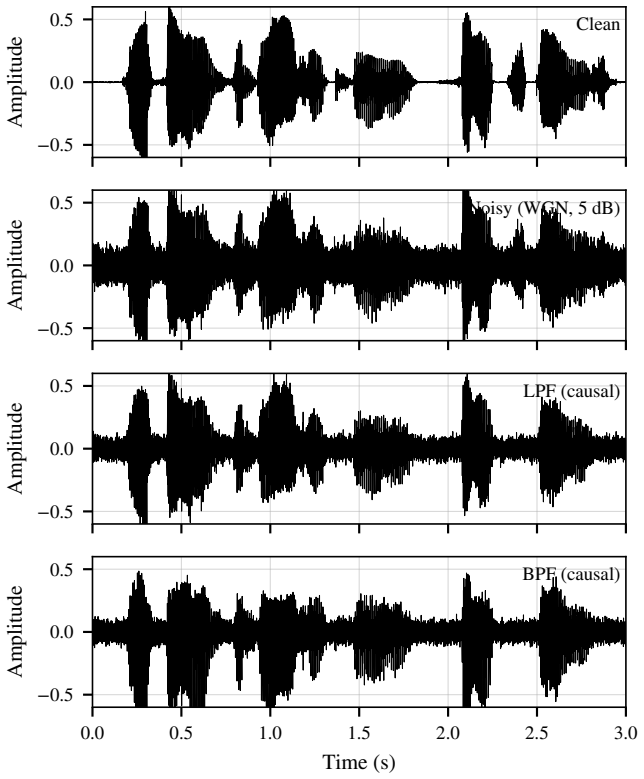


Fig. 4. Clean, noisy (5 dB WGN) and causal LPF/BPF-filtered waveforms (delay-compensated).

has its fundamental and first harmonics, so harmonics are time-shifted relative to higher components. A phase error near 60° is enough for the error energy to equal the signal energy, so waveform SNR collapses even though the magnitude spectrum is nearly preserved. Zero-phase filtering removes this term and turns the same BPF into $+3.15$ dB for 5 dB WGN and $+10.48$ dB (waveform) for rumble. The practical consequence is that offline zero-phase numbers overstate what a real-time system will achieve under a waveform metric, while the output-referred and magnitude metrics show that the causal filter still removes the noise. Which metric matters depends on whether perception of phase distortion matters; conventional objective metrics are known to correlate imperfectly with subjective quality [8], so a listening test is needed to settle this.

Practical guidance. For real-time use, (i) prefer a low-order LPF or a notch bank when the interference is out-of-band or tonal, (ii) avoid HPF/BPF corners close to the speech fundamental unless the phase distortion is acceptable, e.g. lower the corner or use a lower order, (iii) budget buffering plus group delay (Fig. 8b), and (iv) implement filters as biquad cascades with carried state, which here reproduced batch output exactly.

Limitations. The data are six clips from two speakers; the noise is synthetic and stationary; there are no perceptual metrics (PESQ, STOI) or listening tests; cut-off frequencies and orders were fixed a priori rather than optimised per condition; the integer-lag compensation is a coarse proxy; and processing times were measured in a Python sandbox, not on embedded hardware. The ΔSNR of the LPF decreasing from $+3.41$ to $+1.62$ dB as input SNR rises from 0 to 10 dB is consistent with the constant 3.6 dB noise removal being offset by distortion of speech content above 3.4 kHz and by residual phase error, but this attribution was not isolated by a separate experiment.

VI. CONCLUSION

Using real speech and seeded synthetic noise, a causal LPF improved waveform SNR by $+2.89$ dB (5 dB WGN) and

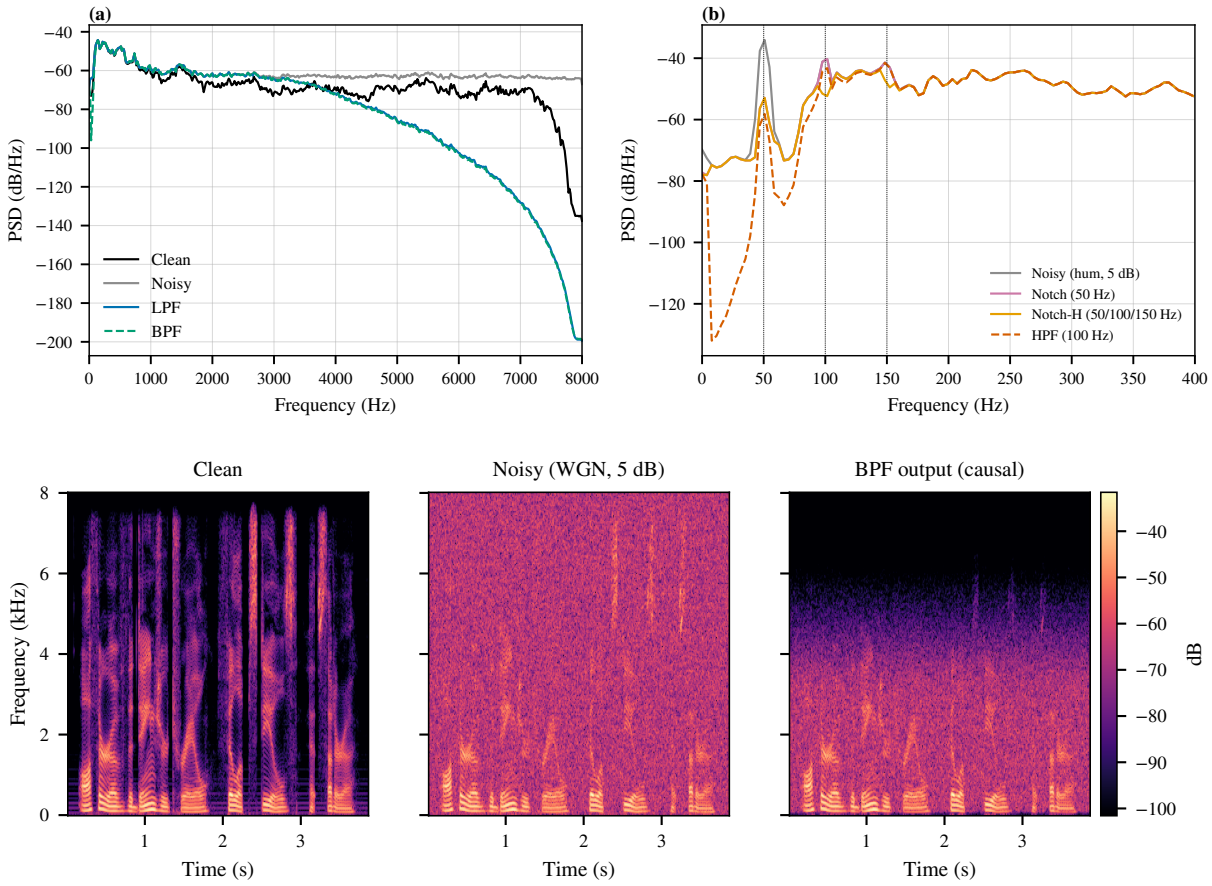


Fig. 5. Frequency-domain views. Top: Welch power spectral densities, (a) 5 dB WGN, full band, and (b) 5 dB hum, 0–400 Hz (dotted lines at 50, 100, 150 Hz). Bottom: spectrograms of the clean signal, the 5 dB WGN mixture and the causal BPF output.

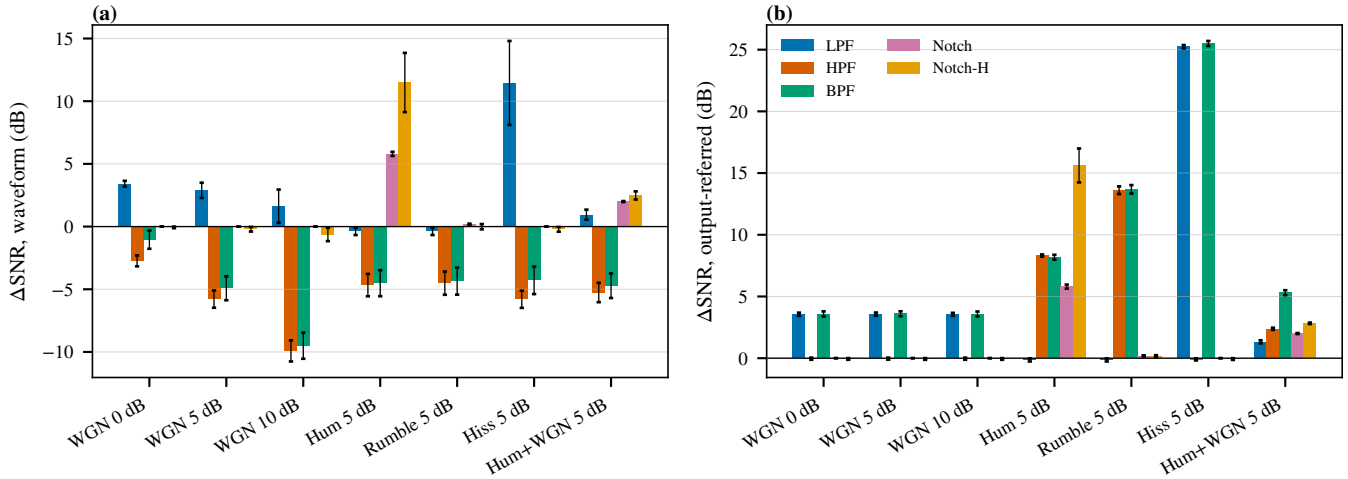


Fig. 6. Causal (delay-compensated) ΔSNR per condition: (a) waveform-referenced, (b) output-referred. Error bars: one standard deviation over 60 trials.

+11.46 dB (hiss), and a three-notch bank by +11.49 dB against 50 Hz hum, while a single notch gave +5.80 dB. Causal HPF and BPF filters removed noise power (up to 13.6 dB output-referred for rumble) but lost waveform SNR through frequency-dependent group delay of about 6 ms near 100 Hz; offline zero-phase filtering hides this cost. Frame-by-frame streaming reproduced batch causal output exactly with sub-millisecond processing time per 10 ms frame in the

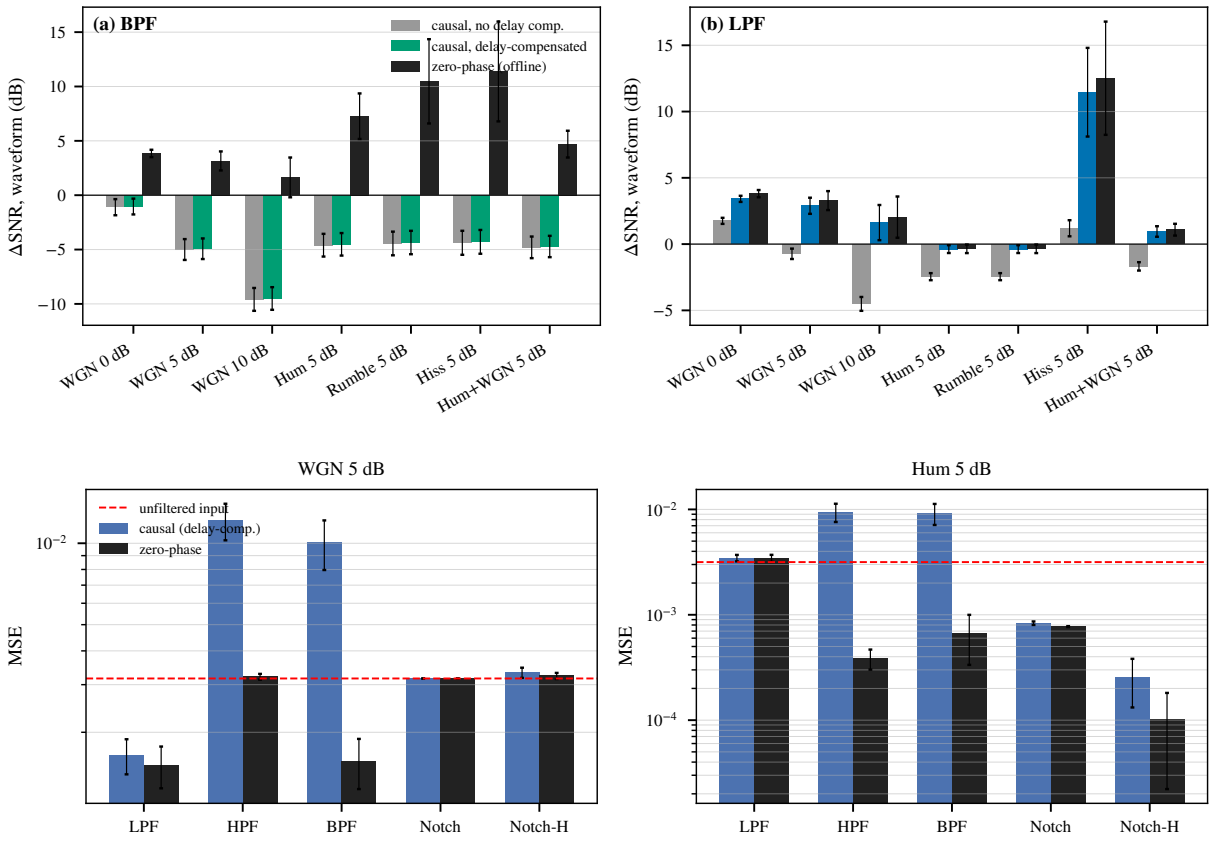


Fig. 7. Top: waveform ΔSNR for raw causal, delay-compensated causal and offline zero-phase filtering, (a) BPF and (b) LPF. Bottom: MSE (log scale) for 5 dB WGN and 5 dB hum, delay-compensated causal versus zero-phase; the dashed line is the unfiltered input.

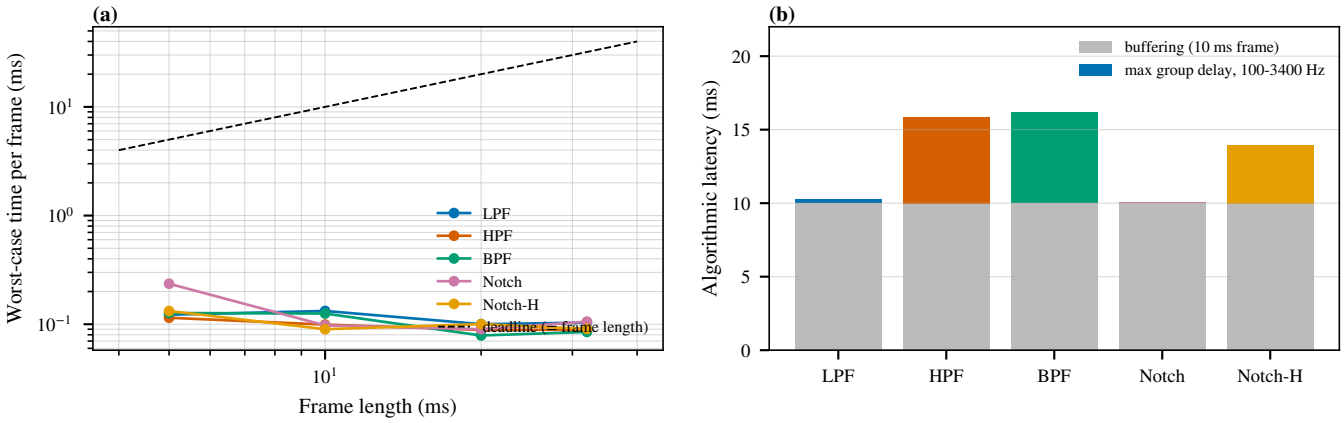


Fig. 8. (a) Worst-case processing time per frame versus frame length. (b) Algorithmic latency: 10 ms buffering plus maximum group delay in 100–3400 Hz (notch centres excluded).

test environment. Future work should add perceptual metrics and listening tests, real recorded noise, adaptive notch/cut-off selection and implementation on embedded targets.

Reproducibility

Code (MATLAB/Octave and Python), the speech clips, per-trial results (`metrics_raw.csv`) and all figures are included in the accompanying archive; see its README.

References

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